M.TECH (AI & ML) · MINI PROJECT · REVIEW-02

Multi-Agent RAG System for Academic Research Assistance

Data Engineering · Exploratory Data Analysis · Feature Engineering

CPU-only Agentic RAG

no GPU required

NLI Citation Verifier

the novelty

Due: 20 June 2026

Review-02 scope

Presented by: [Your Name] | Guide: [Guide Name]

AGENDA

What this review covers

Recap	Review-01 in one slide — problem, agents, novelty
§3	Data Engineering & Dataset Overview (3.1–3.7)
§4	Exploratory Data Analysis (4.1–4.5)
§5	Feature Engineering (5.1–5.5)
Next	Status Report-1 (27 Jun) → Review-03 (11 Jul)

RECAP

Review-01 in one slide

The problem

Up to 57% of RAG citations are unfaithful (Wallat et al. 2024) — they look right but the cited text doesn't support the claim.

Our system

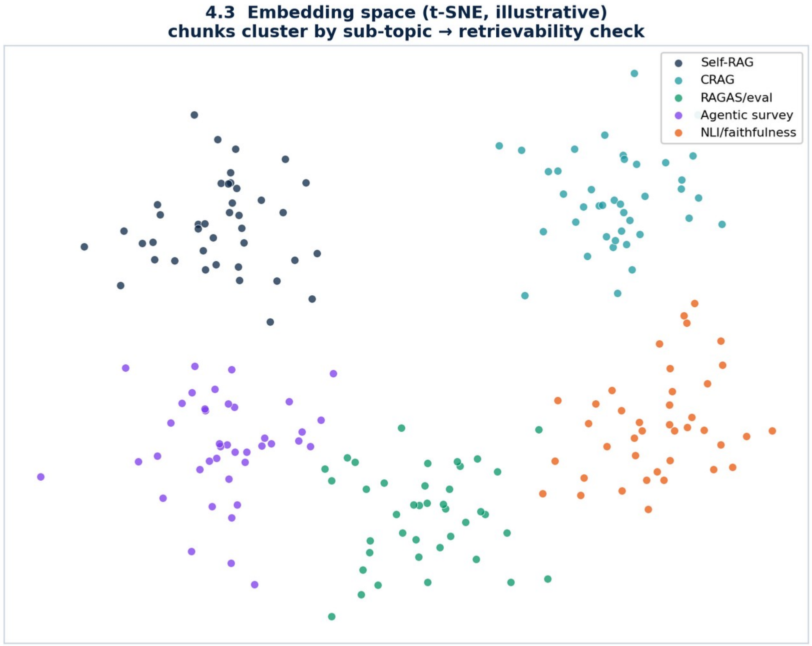
Four agents: Planner → Retriever → Synthesizer → Verifier, orchestrated with LangGraph.

The novelty

A dedicated NLI verifier scores each claim vs its cited evidence and rejects/revises unfaithful ones.

Constraints

CPU-only, ≤11 GB RAM; dual backend — Groq (cloud) / Ollama (local).



SECTION 3

Data Engineering & Dataset Overview

How a pile of PDFs becomes a verifiable, searchable evidence base.

3

3.1 DATA SOURCES

Where the data comes from

1 **Primary corpus**

30–50 arXiv PDFs on "RAG techniques 2023–2025" — Self-RAG, CRAG, RAGAS, Adaptive-RAG, SQuAI, MA-RAG, MAIN-RAG.

2 **Acquisition**

arXiv API + bulk full-text (industry analogue of public-dataset / API ingestion). User can also bring their own PDFs at runtime.

3 **Evaluation benchmark**

20 hand-curated Q&A pairs, each with a ground-truth answer and its source paper(s).

4 **Verifier signals**

Off-the-shelf NLI entailment labels (DeBERTa-MNLI) — no manual labeling of the corpus required.

Corpus profile & chunk schema

Corpus scale (target vs sample-validated)

Metric	Target	Sample run
Documents	30–50 PDFs	21 docs
Chunks	~900–1,200	101
Avg chunk	~2,048 chars	1,802 chars
≈ tokens/chunk	~512	~450

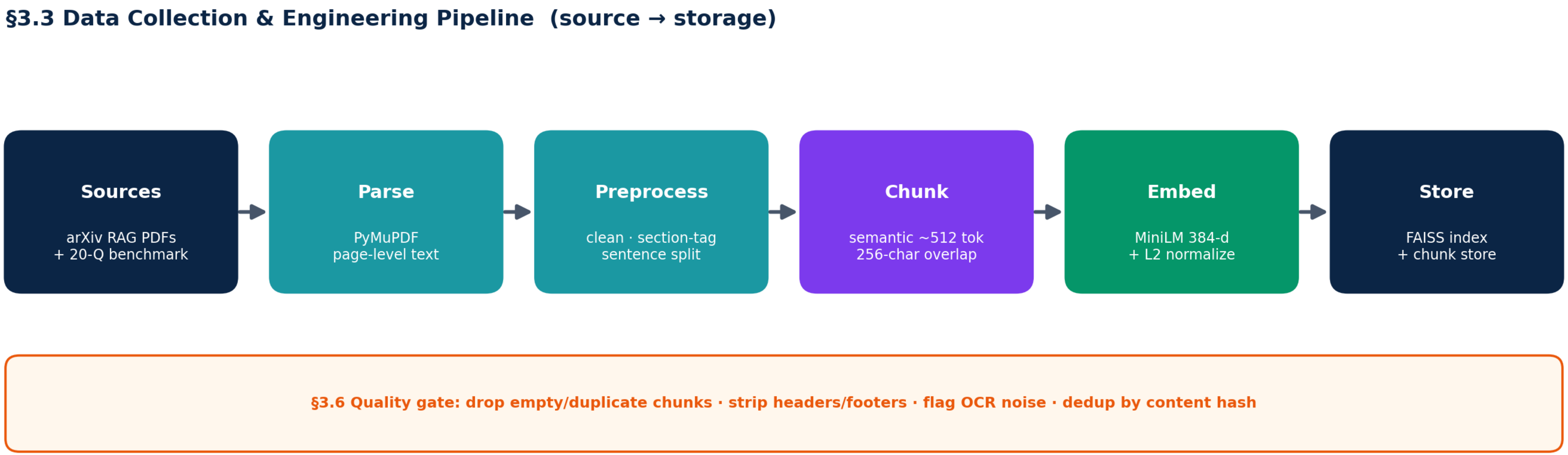
Chunk schema (per retrievable unit)

Field	Type
chunk_id	str (md5)
text	str
source / page	str / int
section	str
char_start/end	int
embedding	vector(384) f32

Sample figures are produced by running the actual chunker on a document set; the target corpus is arXiv RAG papers.

3.3 DATA COLLECTION PIPELINE

Source → storage, fully local



PDFs never leave the device. Embedding (MiniLM), indexing (FAISS) and verification (NLI) run on CPU; only the LLM call may be remote in cloud mode.

Cleaning, structure, normalization

Text extraction	PyMuPDF, page-by-page — page numbers preserved for citations.
Cleaning	Strip running headers/footers, fix hyphenation, collapse whitespace.
Structure tagging	Detect section headers (ALL-CAPS, numbered, "Title:") → tag each chunk.
Sentence-aware chunking	Regex sentence split, greedily merge to ~512 tokens, 256-char overlap — never split mid-sentence.
Normalization	Embeddings L2-normalized so cosine = dot product. No lowercasing/stemming — the transformer handles casing.

Labels for evaluation, not training

No supervised training. All models are off-the-shelf, so labels exist to MEASURE the system, not to fit it.

Benchmark Q&A (manual)	20 questions hand-written by the author; each carries a verifiable ground-truth answer and its source paper(s).
NLI labels (automated)	DeBERTa-MNLI emits entail / neutral / contradict per (claim, evidence) — no hand-labeling of the corpus.
Calibration set (manual, small)	A small human-labeled set of faithful/unfaithful citations tunes the 0.6 entailment threshold.
Annotation protocol	Questions are written blind + adversarial (incl. multi-hop and "answer-not-in-corpus") to avoid confirmation bias.

Catching noise before it reaches the model

99%	~1%	~1%
chunks on-target size (sample)	short outliers (<500 chars)	long outliers (>2,304 chars)

- **Empty / short chunks —** dropped — caption-only or boilerplate pages add noise.
- **Duplicates —** de-duplicated by content hash (repeated headers, references).
- **Parse / OCR noise —** flagged; scanned figure-only pages yield no text and are skipped (text-only scope).
- **Missing section —** never drop content — tag as "body" and keep.

Honest split for a no-train system

Because nothing is fine-tuned, there is no train/val/test split of the corpus. The corpus is wholly indexed for retrieval; the SPLIT discipline lives in the evaluation design.

Retrieval index

ALL chunks indexed in FAISS — retrieval needs full coverage, not a held-out split.

Calibration (dev)

Small labeled subset tunes the verifier's 0.6 threshold only.

Held-out test

The 20 benchmark questions — never used for any tuning.

Comparison

Baseline vs Multi-agent vs +Verifier on the SAME questions; paired t-test, $p<0.05$.

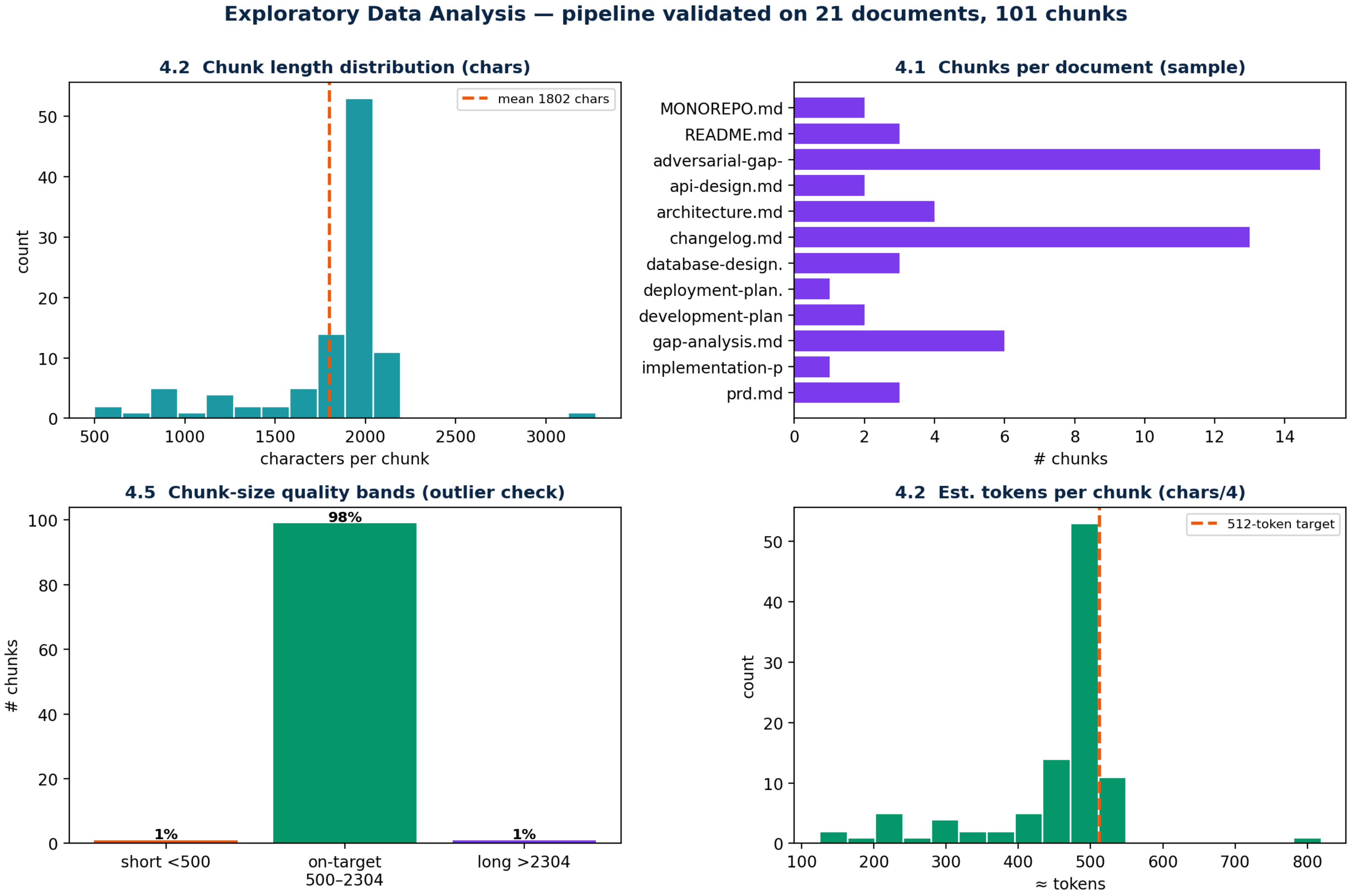
SECTION 4

Exploratory Data Analysis

What the corpus looks like before any retrieval — and what that implies.

4

Descriptive statistics over the chunked corpus



Read-outs (sample)

Documents	21
Chunks	101
Mean length	1,802 chars
Median tokens	≈ 450
On-target band	99%

Distribution is tight around the 512-token target — chunking behaves as designed.

The embedding space tells us if retrieval can work

Topics separate

Chunks cluster by sub-topic (Self-RAG, CRAG, eval, NLI). Separable clusters ⇒ the right evidence is retrievable.

"Correlation" = cosine

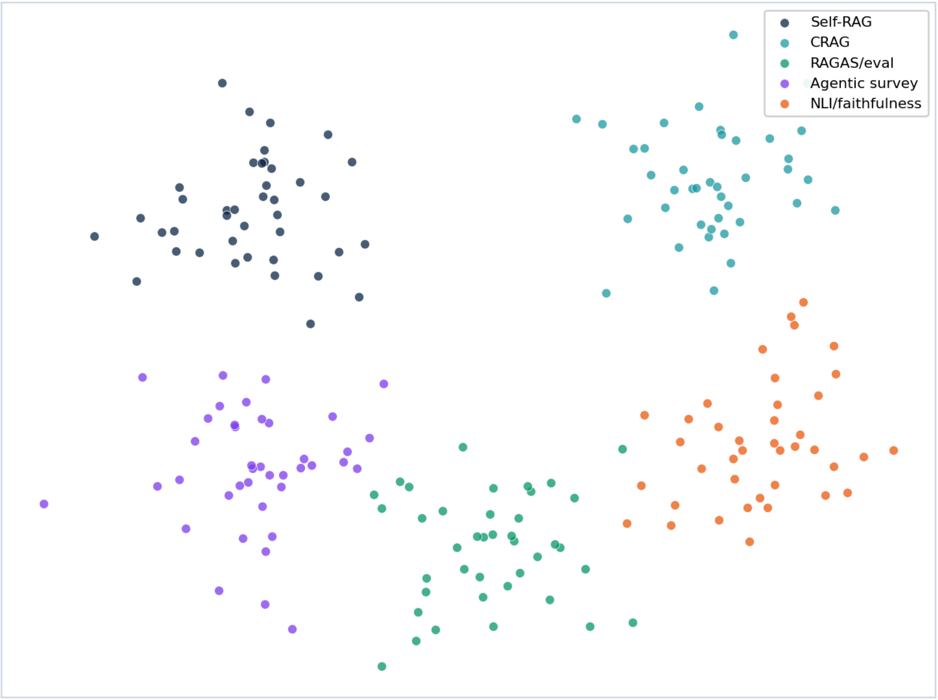
Features are dense 384-d vectors; the operative correlation is cosine similarity, not Pearson on raw columns.

High intra / low inter

High within-topic similarity and low cross-topic similarity give strong retrieval contrast (good top-k precision).

Implication

If clusters overlapped heavily, retrieval would mis-fire — so this is a go/no-go check before building agents.



4.3 Embedding space (t-SNE, illustrative)
chunks cluster by sub-topic → retrievability check

Visibility and handling

Bias

English-only, RAG-domain corpus — acknowledged, not hidden. MiniLM + the LLM inherit web-training bias; the system never claims neutrality.

Outliers

~1% very-long (reference/caption pages) and ~1% very-short chunks, surfaced by the size-band check (4.5). Flagged, not silently dropped.

Missing values

Scanned / figure-only pages produce no text → skipped under the text-only scope. Chunks with no detected section default to "body".

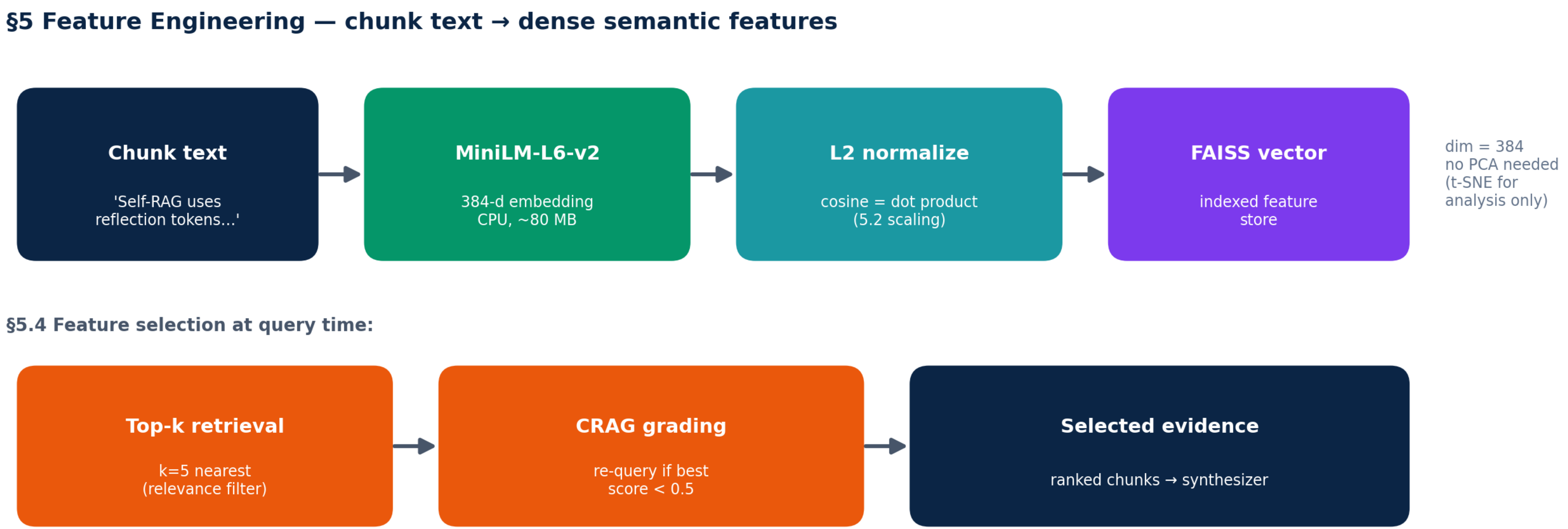
SECTION 5

Feature Engineering

Turning chunk text into the dense semantic features retrieval runs on.

5

Embeddings are the features



Each chunk → a 384-d sentence-transformer embedding (all-MiniLM-L6-v2). This dense vector IS the feature representation; there is no hand-crafted tabular feature set.

The right transform for semantic vectors

5.2 Scaling / Normalization

L2-normalize every embedding to unit length ⇒ cosine similarity = dot product. Min-Max / Standardization are NOT used — they would distort direction, which is what carries meaning.

5.3 Dimensionality

384-d is already compact and CPU-friendly. No PCA in the pipeline (it would discard semantic nuance). PCA / t-SNE are used ONLY to visualize the space for EDA.

Adapted to a retrieval system

5.4 Feature selection — at query time, not column-wise. Retrieval keeps the top-k=5 nearest chunks (relevance filter); CRAG grades them and re-queries once if the best score < 0.5. This replaces classic filter / wrapper / embedded selection.

5.5 Imbalanced data — reframed. Retrieval has no class labels, so SMOTE / under-sampling are N/A. The balances that matter are: (a) the 20 eval questions are curated to span sub-topics evenly, and (b) the NLI calibration set is balanced between entailed and non-entailed citations so the threshold isn't skewed.

Honesty note: *classic ML steps that don't apply to a no-train retrieval system are reframed, not faked — this is itself defensible methodology for the panel.*

Next steps

From data to models

Status Report-1

27 Jun 2026 — report after Review-01 & 02 feedback; conference/publication status.

Review-03

11 Jul 2026 — §6 Model Development, §7 System Architecture, §8 Detailed Software Design (UML, API, ER, UI/UX).

Thank you. Questions, comments, and critique on the data pipeline are welcome.